

Undecidability of complete quantum channel information order

A constructive reduction for rational strictly positive measurements

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Abstract

Complete symmetric logarithmic derivative information order is co-recursively enumerable complete, already for pairs of measurement channels specified by finite strictly positive POVMs with rational complex entries. Its failure is recursively enumerable complete. The same classification holds for unrestricted complete Umegaki relative-entropy order, in which every finite untouched reference and every pair of input states are allowed. There is no total computable bound, from a rational POVM pair, on a reference dimension sufficient to detect every failure of complete SLD order.

The proof reduces the finite-dimensional nonlocal-game promise supplied by $\text{MIP}^* = \text{RE}$ to a uniformly gapped polynomial on the actual two-factor normalized Jordan domain. A quantitative Pauli anchor and local completely positive smoothing establish soundness. An ordered-word graph construction compiles the polynomial to inverse-Jordan kernels, which are embedded into normalized strictly positive POVMs with a dimension-independent global completion. All these reduction steps, and the equivalence between the two channel orders, are proved below. The nonlocal-game hardness theorem is the external complexity premise. A further normalization using its perfect consistent strategies gives a computable positive entropy margin. Finite tensor amplification then proves that the bounded complete entropy defect cannot be uniformly approximated even at the fixed additive error $1/4$.

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1 Definitions and the main result

All Hilbert spaces in individual tests are finite dimensional. Dimensions are not bounded uniformly over tests. Matrices with *rational complex entries* have real and imaginary parts in $\mathbb{Q}$. A faithful state is a positive definite matrix of trace one.

For $W > 0$, write

$$\Omega_W(X) = \frac{WX + XW}{2}, \quad g_W(X) = \langle X, \Omega_W^{-1}(X) \rangle_{\text{HS}}. \quad (1)$$

For Hermitian X , this is the symmetric logarithmic derivative (SLD) quadratic form. In an eigenbasis of W ,

$$g_W(X) = \sum_{i,j} \frac{2|X_{ij}|^2}{w_i + w_j}.$$

We reserve $\mathcal{J}_{U,V}$ for the *unhalved* rectangular Sylvester operator

$$\mathcal{J}_{U,V}(Z) = UZ + ZV. \quad (2)$$

Thus $\Omega_W = \mathcal{J}_{W,W}/2$. This factor of two matters in the POVM construction.

A POVM $E = (E_i)$ on $\mathbb{C}^a$ has $E_i \geq 0$ and $\sum_i E_i = I_a$. Its measurement channel is

$$\mathcal{E}(X) = \sum_i \text{Tr}(E_i X) |i\rangle\langle i|.$$

For an untouched reference R , its output on a state ρ has blocks $\rho_i^E = \text{Tr}_a[(E_i \otimes I_R)\rho]$. Strict positivity means *every* $E_i > 0$.

Definition 1.1. For channels $\mathcal{P}, \mathcal{Q}$ with a common input, write $\mathcal{P} \succeq_{\text{SLD}}^c \mathcal{Q}$ when

$$g_{\mathcal{P}_R(\rho)}(\mathcal{P}_R(H)) \geq g_{\mathcal{Q}_R(\rho)}(\mathcal{Q}_R(H)) \quad (3)$$

for every finite reference R , faithful input state ρ , and traceless Hermitian tangent H , where $\mathcal{P}_R = \mathcal{P} \otimes \text{id}_R$. Write $\mathcal{P} \succeq_{\text{RE}}^c \mathcal{Q}$ when

$$D(\mathcal{P}_R(\alpha) \| \mathcal{P}_R(\beta)) \geq D(\mathcal{Q}_R(\alpha) \| \mathcal{Q}_R(\beta)) \quad (4)$$

for every finite R and every pair of input states α, β . Here $D(A \| B) = \text{Tr } A(\log A - \log B)$ when $\text{supp } A \subseteq \text{supp } B$, and $+\infty$ otherwise, with the usual convention at zero eigenvalues.

In (4) there is **no restriction requiring equal reference marginals**. Output matrices in (3) are restricted to the fixed support of the channel's output on the identity, where they are faithful. For strictly positive POVMs and faithful inputs the output blocks are already positive definite.

Theorem 1.2 (Main theorem). *Consider finite descriptions of pairs of rational complex, strictly positive POVMs on a common finite input space; the input dimension and numbers of outcomes are part of the description.*

1. *The set of pairs (E, F) satisfying $\mathcal{E} \succeq_{\text{SLD}}^c \mathcal{F}$ is coRE-complete under computable many-one reductions. Its complement is RE-complete.*
2. *For every pair of finite-dimensional channels,*

$$\mathcal{P} \succeq_{\text{SLD}}^c \mathcal{Q} \iff \mathcal{P} \succeq_{\text{RE}}^c \mathcal{Q}.$$

Consequently the same computability classification holds for the unrestricted complete Umegaki order on the stated POVM inputs.

3. *There is no total computable function of a valid rational POVM pair that bounds a reference dimension sufficient to witness every failure of its complete SLD order.*

In particular these orders are undecidable, and the set of dominating pairs is not recursively enumerable.

The classification concerns a generic complex class. It does not assert hardness for a transpose-compatible subclass, a fixed input dimension, an equal-reference-marginal entropy order, or numerical contraction coefficients. Every failure in the SLD definition has some finite witness; assertion 3 rules out a *uniform computable sufficient bound* from the input.

1.1 Roadmap and the external premise

The only external hardness premise is the following consequence of Ji, Natarajan, Vidick, Wright and Yuen [1]: from a Turing machine and its input one can effectively construct a finite rational two-player nonlocal game G such that

$$\text{halting} \implies \omega^*(G) = 1, \quad \text{nonhalting} \implies \omega^*(G) \leq \frac{1}{2}.$$

Here ω^* is the supremum over finite-dimensional tensor-product strategies. Attainment of the supremum is not assumed.

Section 2 proves the information-order equivalence. Sections 3–6 encode the game on the actual normalized tensor-Jordan domain and obtain a polynomial with a uniform positive gap in the low-value case and a strict negative witness in the high-value case. Section 7 compiles ordered words to finitely many inverse-Jordan denominators. Sections 8–9 construct legitimate POVMs and control all states and tangents. Section 10 finishes the reduction and the upper computability bound.

Sections 11–14 strengthen the result: the finite nonnegative value $\mathcal{V}(E, F)$ in (64) has no total algorithm for approximation even at the fixed additive tolerance $1/4$. This uses the perfect-consistency part of the external theorem to obtain a witness margin independent of the unknown strategy dimension.

The construction is a finite symbolic algorithm for the output POVMs. It is not an efficient procedure to find an upper-value game's strategy or its witness dimension.

2 Equivalence of unrestricted entropy and SLD order

We prove the equivalence for arbitrary finite-dimensional channels. Absorb any pre-existing untouched reference into the channels whenever convenient.

First, allowing all Hermitian H in (3) is equivalent to the traceless formulation: write $H = H_0 + c\rho$, where $\text{Tr } H_0 = 0$ and $c = \text{Tr } H$. Since $\Omega_A^{-1}(A) = I$ and both channels preserve trace, the added square and cross terms agree after either channel.

2.1 From SLD to relative entropy

For $s > 0$ let $f_s(x) = (x - 1)^2/(x + s)$. For positive definite A, B define

$$D_{f_s}(A\|B) = \langle A - B, (L_A + sR_B)^{-1}(A - B) \rangle_{\text{HS}}, \quad (5)$$

where $L_A(X) = AX$ and $R_B(X) = XB$. In mixed eigenbases this is

$$\sum_{i,j} \frac{(a_i - b_j)^2}{a_i + sb_j} |\langle u_i, v_j \rangle|^2.$$

For faithful states ρ, σ , set $X = \rho - \sigma$ and append a reference qubit:

$$\Theta_s = \frac{1}{1+s} \begin{pmatrix} \rho & 0 \\ 0 & s\sigma \end{pmatrix}, \quad H_X = \begin{pmatrix} 0 & X \\ X & 0 \end{pmatrix}.$$

Solving the off-diagonal Sylvester equations yields

$$g_{\Theta_s}(H_X) = 4(1+s)D_{f_s}(\rho\|\sigma). \quad (6)$$

The same identity holds after either channel. Complete SLD comparison therefore compares every D_{f_s} .

For clarity, the precise positive integral needed here is

$$x \log x - (x-1) = \int_0^\infty \frac{(x-1)^2}{x+s} \frac{s}{(1+s)^2} ds. \quad (7)$$

It follows by elementary integration, or by differentiating twice in x and matching the value and first derivative at $x=1$. For states the affine term cancels, so applying the scalar spectral formula and Tonelli's theorem gives

$$D(A\|B) = \int_0^\infty D_{f_s}(A\|B) \frac{s}{(1+s)^2} ds.$$

This includes the usual infinite support values. Indeed the scalar perspective $h_s(a, b) = (a-b)^2/(a+sb)$, with $h_s(0, 0) = 0$, is continuous on the nonnegative quadrant. Functional calculus for the commuting positive operators L_A, R_B shows that each D_{f_s} is continuous on positive semidefinite pairs in fixed dimension. Common faithful regularization extends its comparison to boundary pairs, and the nonnegative integral then gives the extended-valued entropy comparison. This proves $\succeq_{\text{SLD}}^c \Rightarrow \succeq_{\text{RE}}^c$.

2.2 A saturated reference pair

Suppose now that $\mathcal{P} \succeq_{\text{RE}}^c \mathcal{Q}$. Fix a faithful ρ and Hermitian H on the input including an arbitrary original reference. Introduce another reference with a centre e_0 and N leaves. Choose

$$W = \text{diag}(1, \lambda_1, \dots, \lambda_N), \quad \lambda_k > 0, \quad \tau = vv^*, \quad L = e_0 w^* + w e_0^*, \quad (8)$$

where v, w are leaf-supported and $\|v\| = 1$. Let $Z = \text{Tr } W$ and consider

$$\alpha = \rho \otimes \tau, \quad \beta(\varepsilon) = \frac{\rho \otimes W + \varepsilon H \otimes L}{Z}. \quad (9)$$

The denominator has trace one because $\text{Tr } L = 0$, and it is positive definite for sufficiently small $|\varepsilon|$ of either sign.

For either channel let $A = \Phi(\rho)$ and $K = \Phi(H)$ on its effective support. At $\varepsilon = 0$,

$$D(A \otimes \tau \| A \otimes W/Z) = D(\tau \| W/Z).$$

Thus the source-minus-target entropy difference is nonnegative near zero and vanishes at zero. Its second derivative is nonnegative. Conjugating the auxiliary reference by $\text{diag}(-1, 1, \dots, 1)$ also shows that each entropy along the path is even.

2.3 The denominator Hessian and realizable filters

For $B > 0$ and Hermitian Y , the logarithm resolvent formula gives

$$-\frac{d^2}{d\varepsilon^2} \log(B + \varepsilon Y) \Big|_0 = 2 \int_0^\infty (B + u)^{-1} Y (B + u)^{-1} Y (B + u)^{-1} du. \quad (10)$$

Define the finite resolvent sum

$$F(t) = w^* (W_{\text{leaf}} + tI)^{-1} v.$$

In an eigenbasis of A , the two exterior resolvents in (10) act on leaves, and the middle resolvent acts on the centre. For the ordered system indices (i, j) the contribution is

$$2|K_{ij}|^2 a_i \int_0^\infty \frac{|F(u/a_i)/a_i|^2}{a_j + u} du.$$

Substituting $u = a_i t$ proves the exact identity

$$\frac{d^2}{d\varepsilon^2} D \left(A \otimes \tau \left\| \frac{A \otimes W + \varepsilon K \otimes L}{Z} \right\| \right) \Big|_0 = 2 \sum_{i,j} |K_{ij}|^2 \int_0^\infty \frac{|F(t)|^2}{a_j + a_i t} dt. \quad (11)$$

The normalization contributes only $\log Z$ and does not affect the derivative. The integrals converge because F is bounded at zero and $O(t^{-1})$ at infinity.

Every finite simple-resolvent sum $F(t) = \sum_k c_k / (t + \lambda_k)$ is realizable: take $v_k = N^{-1/2}$ and $w_k = \sqrt{N} \bar{c}_k$. Large but finite coefficients merely make the admissible interval of ε smaller.

Lemma 2.1 (Finite filters concentrating at one). *For $m \geq 1$ define $\lambda_j = 1 + j/m^3$ for $0 \leq j \leq 2m$ and*

$$F_m(t) = \frac{t^m}{\prod_{j=0}^{2m} (t + \lambda_j)}, \quad I_m = \int_0^\infty F_m(t)^2 dt.$$

These functions are realizable finite resolvent sums with rational poles and coefficients, and for $a, b > 0$,

$$\frac{1}{I_m} \int_0^\infty \frac{F_m(t)^2}{b + at} dt \rightarrow \frac{1}{a + b}. \quad (12)$$

Proof. Partial fractions give

$$F_m(t) = \sum_{j=0}^{2m} \frac{c_j}{t + \lambda_j}, \quad c_j = \frac{(-\lambda_j)^m}{\prod_{k \neq j} (\lambda_k - \lambda_j)} \in \mathbb{Q}.$$

Put $f_m(t) = t^m / (1 + t)^{2m+1}$. Since $\log(1 + x) \leq x$,

$$\exp(-2/m - 1/m^2) \leq F_m(t)/f_m(t) \leq 1 \quad (t > 0).$$

The normalized density proportional to f_m^2 is beta-prime with parameters $(2m + 1, 2m + 1)$. Direct beta-integral evaluation gives its mean $(2m + 1)/(2m)$ and variance

$$\frac{(2m + 1)(4m + 1)}{(2m - 1)(2m)^2}.$$

It therefore converges in probability to one. The displayed multiplicative bounds show that the normalized F_m^2 densities differ from these densities by a total-variation error tending to zero. Applying this to the bounded continuous function $(b + at)^{-1}$ proves (12). $\square$

Apply the entropy hypothesis to (9) with each F_m . The difference of (11) is nonnegative. Divide by I_m and take the limit, using the finitely many output eigenvalues. The result is exactly $g_{\mathcal{P}(\rho)}(\mathcal{P}(H)) \geq g_{\mathcal{Q}(\rho)}(\mathcal{Q}(H))$. The original reference was arbitrary, proving $\succeq_{\text{RE}}^c \Rightarrow \succeq_{\text{SLD}}^c$. Every auxiliary reference used has finite dimension $2m + 2$. If one assumes entropy comparison only for faithful numerators, replace τ by $(1 - \zeta)\tau + \zeta I/(N + 1)$ and pass to $\zeta \downarrow 0$; saturation persists and the Hessian is linear and continuous in the numerator.

3 The actual normalized tensor Jordan domain

Fix physical dimension h . For a Hermitian physical effect A and $R > 0$ on $\mathbb{C}^h \otimes \mathbb{C}^n$, write

$$\mathcal{A}_A(R) = \text{Tr}_h[(A \otimes I_n)R].$$

For $S > 0$ on $\mathbb{C}^h \otimes \mathbb{C}^m$ define, in a fixed tensor-vectorization convention,

$$B_A(R, S) = \mathcal{A}_A(R) \otimes I_m + I_n \otimes \mathcal{A}_A(S)^\top, \quad D = B_I(R, S), \quad Z_A = D^{-1/2} B_A D^{-1/2}. \quad (13)$$

The notation $\mathcal{A}_A(S)^\top$ means $(\mathcal{A}_A(S))^\top$. Tensor-factor swaps turn this convention into ordinary column vectorization of (2); nothing depends on that fixed swap. If $\alpha I \leq A \leq (1 - \alpha)I$, then

$$\alpha I \leq Z_A \leq (1 - \alpha)I, \quad B_{I-A} = D - B_A. \quad (14)$$

All effects in a tuple use the *same* R, S, D . We never enlarge this domain to arbitrary unrelated operator tuples.

3.1 The physical Pauli and payload algebra

Pad the game to M questions and K outcomes on each side. Zero-probability questions and duplicated outcomes preserve its value. Put $N = MK$, $h = 2N$, and let e_g , $g = (x, a)$, be the minimal projections of $\mathbb{C}^N$. Inside M_h use $M_2 \otimes \mathbb{C}^N$. With the usual Pauli matrices $\sigma_X, \sigma_Y, \sigma_Z$ and $\tau = 1/2$, take

$$A_i = \frac{I + \tau \sigma_i \otimes I}{2}, \quad A_{i,g} = \frac{I + \tau \sigma_i \otimes e_g}{2}. \quad (15)$$

These rational complex effects and their complements lie between $I/4$ and $3I/4$. Define centered contractions

$$T_i = (2Z_{A_i} - I)/\tau, \quad T_{i,g} = (2Z_{A_{i,g}} - I)/\tau. \quad (16)$$

They are selfadjoint contractions by positivity of the normalized maps.

Put $W = \text{Tr}_h R$, $V = (\text{Tr}_h S)^\top$, and on the two reference factors set

$$r = W \otimes I, \quad s = I \otimes V, \quad p = r^{1/2}(r+s)^{-1/2}, \quad q = s^{1/2}(r+s)^{-1/2}, \quad j = (r-s)(r+s)^{-1}. \quad (17)$$

The Choi correspondence provides UCP maps of the physical algebra. Explicitly, the left map is $B \mapsto W^{-1/2} \text{Tr}_h[(B^\top \otimes I)R]W^{-1/2}$; the right map is $B \mapsto V^{-1/2}(\text{Tr}_h[(B \otimes I)S])^\top V^{-1/2}$. Both are completely positive and unital. The left Pauli orientation is therefore transposed and the right orientation is not.

After common local Stinespring dilations of the *entire* algebra, write the exact opposite Pauli representations as x_i, y_i , and the local payload projections as a_g, b_g . They commute with their respective Pauli representations and form global POVMs. The normalized variables have the form

$$T_i = p x_i p + q y_i q, \quad T_{i,g} = p x_i a_g p + q y_i b_g q. \quad (18)$$

For general UCP compressions this formula is used through the positive-weight dilation limit proved below. It is not a claim that arbitrary decomposable compressions satisfy the same tensor-local relations.

3.2 Honest finite strategy realization

Given local question-wise POVMs $M_{a|x}$ and $N_{b|y}$, set $a_{x,a} = M_{a|x}/M$, $b_{y,b} = N_{b|y}/M$. Define UCP maps on $\bigoplus_{g=1}^N M_2$ by

$$\Phi_A((Z_g)_g) = \sum_g Z_g \otimes a_g, \quad \Phi_B((Z_g)_g) = \sum_g Z_g \otimes b_g.$$

Their Choi baselines have identity reference marginals. Thus $p = q = I/\sqrt{2}$ and

$$T_i = (x_i + y_i)/2, \quad T_{i,g} = (x_i a_g + y_i b_g)/2, \quad x_i = \sigma_i^\top, \quad y_i = \sigma_i, \quad (19)$$

with the strategy-space tensor factors understood. The baselines may be singular, but $D = 2I$ remains invertible. These honest models belong to the positive semidefinite closure of the domain. Strict negativity will later be preserved by an explicit faithful perturbation.

4 A quantitative augmented Pauli anchor

For exact opposite Pauli representations in (18) and any vector v , set

$$\begin{aligned} e_i(v) &= \langle v, (I - T_i^2)v \rangle, & e(v) &= \sum_i e_i(v), \\ c(v) &= \sum_{(i,j) \text{ cyclic}} \|[T_i, T_j]v\|^2, & \delta(v) &= e(v) + c(v). \end{aligned} \quad (20)$$

Let $P = |\Phi_2\rangle\langle\Phi_2| \otimes I$ be the Bell projector for the two opposite Pauli representations, and $Q = I - P$.

Lemma 4.1 (Anchor rigidity). *For every vector v ,*

$$\|Qv\|^2 \leq 10\delta(v), \quad \|jv\|^2 \leq 10\delta(v), \quad \|(p - 2^{-1/2})v\|^2, \|(q - 2^{-1/2})v\|^2 \leq 5\delta(v). \quad (21)$$

Also

$$\|T_i v - x_i v\| \leq 7\sqrt{\delta(v)}, \quad \|jT_i v\| \leq 2\sqrt{\delta(v)}. \quad (22)$$

The same estimates hold for arbitrary UCP maps after fixed common local Stinespring isometries, independently of dimensions and marginal condition numbers.

Proof. Write $L_i = px_i p$, $R_i = qy_i q$, and $U_i = L_i - R_i$. The column $C : v \mapsto (pv, qv)$ is an isometry, and $T_i = C^* \pi_i C$ for $\pi_i = x_i \oplus y_i$. The residual $\rho_i = \pi_i C v - C T_i v$ has norm squared $e_i(v)$. Opposite chirality gives, for cyclic (i, j, k) ,

$$\|[T_i, T_j]v + 2iU_k v\| \leq \sqrt{e_i(v)} + \sqrt{e_j(v)}. \quad (23)$$

Indeed compare $T_i T_j v$ with $C^* \pi_i \pi_j C v$ and reverse i, j . Hence

$$\sum_k \|U_k v\|^2 \leq \frac{3}{4}c(v) + \frac{3}{2}e(v). \quad (24)$$

The same comparison using the anticommutator gives $\|\{T_i, T_j\}v\| \leq \sqrt{e_i(v)} + \sqrt{e_j(v)}$ for $i \neq j$. The conditioning-free step is an inverse-root identity. Put

$$A = r^{-1}, \quad B = s^{-1}, \quad F = A+B, \quad a = A^{1/2}, \quad b = B^{1/2}, \quad \xi = F^{-1/2}v, \quad \alpha = a\xi, \quad \beta = b\xi.$$

All functions of A, B commute, while $[b, x_i] = [a, y_i] = 0$. The orthogonal complement column to C is $(q, -p)$, so

$$e_i(v) = \|d_i\|^2, \quad d_i = F^{-1/2}ab(x_i - y_i)\xi, \quad U_i v = F^{-1/2}(Bx_i - Ay_i)\xi.$$

Adding these formulas gives the exact identity

$$x_i\beta - y_i\alpha = F^{1/2}(a+b)^{-1}(U_i v + d_i). \quad (25)$$

Its multiplier has norm at most one because $\sqrt{a^2 + b^2}/(a+b) \leq 1$ in commuting functional calculus. Therefore

$$E_* := \sum_i \|x_i\beta - y_i\alpha\|^2 \leq 2 \sum_i (\|U_i v\|^2 + e_i(v)) \leq 5\delta(v).$$

The commuting involutions $c_i = x_i y_i$ satisfy $\sum_i c_i = 4P - I$. Decomposing into P, Q gives

$$E_* = 3\|P(\alpha - \beta)\|^2 + 3\|Q\alpha\|^2 + 3\|Q\beta\|^2 + 2\operatorname{Re}\langle Q\alpha, Q\beta \rangle.$$

It follows that

$$\|\alpha - \beta\|^2 \leq E_*, \quad \|Q\alpha\|^2 + \|Q\beta\|^2 \leq E_*/2.$$

The scalar inequality $|\sqrt{a^2 + b^2} - (a+b)/\sqrt{2}| \leq |a-b|/\sqrt{2}$ shows $\|v - (\alpha + \beta)/\sqrt{2}\| \leq \|\alpha - \beta\|/\sqrt{2}$. Thus $\|Qv\| \leq \sqrt{2E_*}$. Moreover

$$jv = (b-a)(b+a)F^{-1/2}\xi, \quad \|jv\| \leq \sqrt{2}\|\alpha - \beta\| \leq \sqrt{2E_*}.$$

This proves the first two bounds in (21); the others use $|\sqrt{(1+t)/2} - 1/\sqrt{2}| \leq |t|/\sqrt{2}$ for $-1 \leq t \leq 1$.

For the action bounds write the components of ρ_i as $a'_i = x_i p v - p T_i v$, $b'_i = y_i q v - q T_i v$. Then

$$U_i v = j T_i v + p a'_i - q b'_i, \quad \|j T_i v\| \leq \|U_i v\| + \sqrt{e_i(v)} \leq 2\sqrt{\delta(v)}.$$

Expanding $T_i v - x_i v$ with these residuals and the preceding scalar weight estimates gives

$$\|T_i v - x_i v\| \leq \|U_i v\| + \|jv\| + (1 + \sqrt{2})\sqrt{e_i(v)} \leq 7\sqrt{\delta(v)}.$$

Finally let J_H, J_K be fixed local Stinespring isometries for the complete physical algebra. On the enlarged left space extend the weight to $W_\varepsilon = J_H W J_H^* + \varepsilon(I - J_H J_H^*)$, and likewise on the right. For every $\varepsilon > 0$ the exact-representation proof applies. On the embedded original tensor product, D_ε restricts to the original D , and the compression of each normalized letter is the original letter. Off-subspace entries applied to an original vector tend to zero: they contain a factor $\sqrt{\varepsilon}$, while the other original marginal is bounded below for the fixed instance. Uniform boundedness of the letters gives convergence of every fixed finite word. Their energies converge, the local isometries and P remain fixed, and the balance operator on the original subspace is unchanged. Taking $\varepsilon \downarrow 0$ proves the claimed UCP version. No uniform convergence rate is required. $\square$

4.1 Transport of words and a single payload

Define honest balanced anchors $K_i = (x_i + y_i)/2$. On a Bell vector, $K_i P v = x_i P v$ and $K_i x_j P v = \delta_{ij} P v$. The commutator and anticommutator residual bounds imply, for $i \neq j$,

$$\|T_i T_j v\| \leq \frac{\|[T_i, T_j]v\| + \sqrt{e_i(v)} + \sqrt{e_j(v)}}{2} \leq \sqrt{\delta(v)}.$$

For $i = j$, $\|(T_i^2 - I)v\| \leq \sqrt{e_i(v)}$. Reduce pairs from the right in a word w of length d . Each replacement costs at most $\sqrt{\delta(v)}$, and all unreduced prefixes are contractions. The corresponding K word makes exactly the same reductions on Pv . Consequently

$$\|T_w v - K_w P v\| \leq (\lfloor d/2 \rfloor + \sqrt{10} + 7\mathbf{1}_{d \text{ odd}})\sqrt{\delta(v)}. \quad (26)$$

The same reductions and (22) give

$$\|jT_w v\| \leq b_d \sqrt{\delta(v)}, \quad b_d = \lfloor d/2 \rfloor + \sqrt{10} + 2. \quad (27)$$

Fix a local payload effect a_g, b_g , and put $K_{H,g} = (x_H a_g + y_H b_g)/2$. Moving the outside p, q factors to the bra in $\langle T_u w, T_{H,g} T_t v \rangle$, replacing them by $1/\sqrt{2}$ using (27), and then using (26), proves, for $|u|, |t| \leq d$,

$$\begin{aligned} & |\langle T_u w, T_{H,g} T_t v \rangle - \langle K_u P w, K_{H,g} K_t P v \rangle| \\ & \leq C_d (\|w\| \sqrt{\delta(v)} + \|v\| \sqrt{\delta(w)}), \quad C_d = \sqrt{2} b_d + \lfloor d/2 \rfloor + \sqrt{10} + 7. \end{aligned} \quad (28)$$

This proof uses only local contractions and tested word actions.

Choose cyclic indices $(K, L, H) = (X, Y, Z)$ and define Hermitian extractors

$$\begin{aligned} L_H &= T_H + i[T_K, T_L], & R_H &= T_H - i[T_K, T_L], \\ U_g &= T_{H,g} + i[T_K, T_{L,g}], & V_g &= T_{H,g} - i[T_K, T_{L,g}], \\ P_g &= (L_H U_g + U_g L_H)/2, & Q_g &= (R_H V_g + V_g R_H)/2. \end{aligned} \quad (29)$$

They have degree at most four, operator norm at most nine, and monomial coefficient absolute sum at most nine. Every monomial has exactly one raw payload occurrence. In the honest balanced model, opposite chirality gives $L_H = x_H$, $R_H = y_H$, $U_g = x_H a_g$, $V_g = y_H b_g$, so $P_g = a_g$, $Q_g = b_g$ as operators.

Since a_g, b_g commute with the Bell projector, (28) removes the Bell projections at an extra cost at most $\sqrt{10}\|w\|\sqrt{\delta(v)}$. Using $d = 3$ and $9C_3 + \sqrt{10} < 200$ gives

$$|\langle w, (P_g - a_g)v \rangle| \leq 200(\|w\|\sqrt{\delta(v)} + \|v\|\sqrt{\delta(w)}), \quad (30)$$

and the same estimate for $Q_g - b_g$. This is initially a weak matrix-element estimate, not yet an action-norm estimate.

5 Local completely positive smoothing

The next argument promotes (30) to local action approximation while imposing only finitely many good-descendant conditions.

5.1 A smoothing contraction from local unitaries

For strictly positive exact-representation weights define

$$\Lambda = \log r - \log s, \quad G = e^{-\Lambda^2/2}, \quad x_0 = I, \quad M_0 = \frac{P + \sum_{i=0,X,Y,Z} x_i G x_i}{5}. \quad (31)$$

It is a positive contraction and an average of product-local unitaries. To see the latter, $P = (I + \sum_i x_i y_i)/4$ is such an average, and the Gaussian Fourier identity expresses G as

the average of $r^{it}s^{-it}$ with a standard normal parameter t . Conjugating by x_i preserves the property. Hence for every integer $n \geq 1$,

$$M_0^n = \mathbb{E}_\omega[U_\omega \otimes V_\omega] \quad (32)$$

for local unitaries on the two finite spaces. This average is an analytic existence device; it is not part of the symbolic output.

Set

$$E_M(v) = \langle v, (I - M_0)v \rangle, \quad E_0(v) = \|Qv\|^2 + \sum_{i=0,X,Y,Z} \|jx_i v\|^2.$$

Since $j = \tanh(\Lambda/2)$, the scalar bounds

$$\tanh^2(t/2) \leq 1 - e^{-t^2/2} \leq 8 \tanh^2(t/2)$$

imply

$$E_0(v)/5 \leq E_M(v) \leq 8E_0(v)/5. \quad (33)$$

For the lower scalar bound use $\cosh(t/2) \leq e^{t^2/8}$; for the upper use $\tanh(|t|/2) \geq |t|/4$ when $|t| \leq 1$ and $\tanh(1/2)^2 > 1/5$ otherwise.

Lemma 5.1 (Energy comparison). *Uniformly over all the exact-representation instances,*

$$E_M(v) \leq 421\delta(v), \quad \delta(v) \leq 3000E_M(v). \quad (34)$$

Therefore

$$\delta(M_0^n w) \leq \frac{3000}{2n+1} \|w\|^2, \quad \|v - M_0^n v\| \leq \sqrt{nE_M(v)}. \quad (35)$$

Proof. By Lemma 4.1, the Q and j terms cost at most $10\delta(v)$ each, and $\|jx_i v\| \leq \|jT_i v\| + \|T_i v - x_i v\| \leq 9\sqrt{\delta(v)}$. Thus $E_0 \leq 263\delta$, yielding the first bound.

For the reverse bound put $h_i = \|jx_i v\|$, $h_0 = \|jv\|$, and $b = \|Qv\|$. The exact compression defect is $d_i = qx_i p v - py_i q v$, with $\|d_i\|^2 = e_i(v)$. Replacing p, q by $1/\sqrt{2}$ using the scalar weight estimates, using $\|(x_i - y_i)v\| \leq 2b$ and the cross-party commutations, gives

$$\|d_i\|, \|U_i v\| \leq \sqrt{2}h_0 + h_i + 2b \leq \sqrt{7E_0(v)}.$$

Equation (23) gives $\|[T_i, T_j]v\| \leq 4\sqrt{7E_0(v)}$. Hence $\delta(v) \leq 357E_0(v) \leq 1785E_M(v)$, and the stated constant 3000 is sufficient. Finally the scalar inequalities $t^{2n}(1-t) \leq 1/(2n+1)$ and $(1-t^n)^2 \leq n(1-t)$ on $[0, 1]$, applied to M_0 , prove (35). $\square$

5.2 From weak transport to local action transport

Use the same averaging (32) for every payload on its party:

$$a_g^{(n)} = \mathbb{E}_\omega U_\omega a_g U_\omega^*, \quad b_g^{(n)} = \mathbb{E}_\omega V_\omega b_g V_\omega^*. \quad (36)$$

These are local positive contractions and each family sums to the identity. Applying (30) with w replaced by $M_0^n w$ and using (35) gives by duality

$$\|M_0^n (P_g - a_g)v\| \leq 200\sqrt{\delta(v)} + 200\sqrt{3000/(2n+1)}\|v\|.$$

Also the exact average identity

$$\mathbb{E}_\omega \|(U_\omega \otimes V_\omega)v - v\|^2 = 2\langle v, (I - M_0^n)v \rangle \leq 2nE_M(v)$$

shows $\|M_0^n a_g v - a_g^{(n)} v\| \leq \sqrt{2n E_M(v)}$. Combining with (35) on $P_g v$ proves

$$\begin{aligned} \|P_g v - a_g^{(n)} v\| &\leq \sqrt{421n\delta(P_g v)} + \sqrt{842n\delta(v)} \\ &\quad + 200\sqrt{\delta(v)} + 200\sqrt{3000/(2n+1)}\|v\|. \end{aligned} \quad (37)$$

The same estimate holds for $Q_g, b_g^{(n)}$. For a unit root z let

$$E(z) = \delta(z) + \sum_g \delta(P_g z) + \sum_g \delta(Q_g z). \quad (38)$$

For every $n \geq 1$, each action error in (37) is at most

$$\kappa_n = C_s(\sqrt{n}\sqrt{E(z)} + n^{-1/2}), \quad C_s = 12000. \quad (39)$$

This follows directly by bounding the four coefficients in (37); it requires no small-energy assumption.

5.3 Question normalization without an invertibility assumption

Although $a_g^{(n)}$ form a global POVM, the scaled effects $Ma_{x,a}^{(n)}$ need not sum to I for each question. Put

$$C_x = \sum_a Ma_{x,a}^{(n)}, \quad D_y = \sum_b Mb_{y,b}^{(n)}.$$

They satisfy $0 \leq C_x, D_y \leq MI$. Their square-root measurement columns have norms at most $\sqrt{M}$. For one such column C with $C^*C = Q$, extend its polar partial isometry to an isometry N by filling $\ker Q$ in one chosen outcome. This is possible within the same column space because all $\sqrt{E_a}$ annihilate $\ker Q$. Then

$$C = N\sqrt{Q}, \quad \|(C - N)z\| = \|(\sqrt{Q} - I)z\| \leq \|(Q - I)z\|.$$

The POVM of N has effects $Q^{-1/2}E_a Q^{-1/2}$ on $\text{supp } Q$, with the kernel added to one outcome. It is a genuine local POVM even if Q has a kernel.

Let $\tau_x = \|(C_x - I)z\|$ and $v_y = \|(D_y - I)z\|$. The difference of the joint unnormalized and normalized columns at questions x, y has norm at most $\sqrt{M}v_y + \tau_x$. The first joint column has norm at most M and the second is an isometry. For any diagonal payoff contraction, its expectation therefore changes by at most $(M+1)(\sqrt{M}v_y + \tau_x)$. Averaging with the game distribution π and applying Cauchy-Schwarz gives a score error at most

$$C_3\sqrt{H_n}, \quad C_3 = (M+1)^2, \quad H_n = \sum_x \pi_A(x)\tau_x^2 + \sum_y \pi_B(y)v_y^2. \quad (40)$$

Here we used $(M+1)\sqrt{M+1} \leq C_3$. No scaled effect $Ma_g^{(n)}$ has been incorrectly treated as a contraction.

5.4 Passage back to arbitrary UCP maps

For original compressed maps, extend the weights by εI on the Stinespring complements as in Lemma 4.1. Perform the entire smoothing and measurement rounding on those finite enlarged spaces with $\varepsilon > 0$. Every resulting strategy is a valid finite-dimensional strategy and so is bounded by $\omega^*(G)$. For each fixed original instance all finitely many word expectations, descendant energies and normalization errors converge as $\varepsilon \downarrow 0$. Pass the resulting scalar score inequality to the limit. All displayed constants are independent of ε and conditioning. No compactness across dimensions, maximizing strategy, or uniform rate of this auxiliary limit is needed.

6 A finite game penalty with a uniform gap

Let G have rational question distribution π and rational payoff $0 \leq V(a, b|x, y) \leq 1$. Define the positive polynomial

$$\Delta = \sum_i (I - T_i^2) + \sum_{(i,j) \text{ cyclic}} [T_i, T_j]^* [T_i, T_j].$$

Its quadratic form is δ . Define

$$\begin{aligned} E_{\text{op}} &= \Delta + \sum_g P_g \Delta P_g + \sum_g Q_g \Delta Q_g, \\ H_{\text{op}} &= \sum_x \left(M \sum_a P_{x,a} - I \right)^2 + \sum_y \left(M \sum_b Q_{y,b} - I \right)^2, \end{aligned} \quad (41)$$

and put $E(z) = \langle z, E_{\text{op}} z \rangle$, $H(z) = \langle z, H_{\text{op}} z \rangle$. The first has degree at most twelve, the second at most eight. Both are positive on the actual domain. The Hermitian game-score polynomial is

$$B = M^2 \sum_{x,y,a,b} \pi(x, y) V(a, b|x, y) \frac{P_{x,a} Q_{y,b} + Q_{y,b} P_{x,a}}{2}. \quad (42)$$

In the honest model (19), on a Bell anchor tensored with any strategy vector, every extracted descendant keeps its Bell factor. Thus $E = H = 0$, and $\langle z, Bz \rangle$ is exactly the strategy's game score.

For an arbitrary actual instance and unit z , the two action approximations with error κ_n and $\|P_g\|, \|Q_h\| \leq 9$ show

$$|\langle P_g z, Q_h z \rangle - \langle a_g^{(n)} z, b_h^{(n)} z \rangle| \leq 10\kappa_n.$$

There are K^2 outcome pairs for each question pair. The discrepancy between (42) and the unnormalized local score is at most $10M^2 K^2 \kappa_n$. Moreover Minkowski's inequality gives

$$\sqrt{H_n} \leq \sqrt{H} + \sqrt{2} M K \kappa_n \leq \sqrt{H} + 2 M K \kappa_n.$$

Use (40) to normalize the questions. The resulting valid local strategy has score at most $\omega^*(G)$, so

$$\langle z, Bz \rangle \leq \omega^*(G) + C_1 C_s \sqrt{n} \sqrt{E} + C_1 C_s / \sqrt{n} + C_3 \sqrt{H}, \quad C_1 = 10M^2 K^2 + 2C_3 M K. \quad (43)$$

The preceding dilation argument proves this on every actual normalized domain instance.

Proposition 6.1 (Promised gapped polynomial). *Given rational r and $\gamma > 0$, one can effectively construct a rational Hermitian polynomial F of degree at most twelve such that*

1. *If $\omega^*(G) \leq r - \gamma$, then $F(T) \geq (\gamma/2)I$ throughout the actual normalized tensor-Jordan domain.*
2. *If $\omega^*(G) > r + \gamma/2$, then F has a strict negative finite-dimensional witness with faithful baselines.*

Proof. Choose

$$\begin{aligned} s_0 &= \max\{1, \lceil 4C_1 C_s / \gamma \rceil\}, & n &= s_0^2, & C_2 &= C_1 C_s s_0, \\ \lambda_E &= \lceil 2C_2^2 / \gamma \rceil, & \lambda_H &= \lceil 2C_3^2 / \gamma \rceil, \end{aligned}$$

and set

$$F = rI - B + \lambda_E E_{\text{op}} + \lambda_H H_{\text{op}}. \quad (44)$$

In the low-value case, (43) gives

$$\langle z, Fz \rangle \geq 3\gamma/4 - C_2\sqrt{E} - C_3\sqrt{H} + \lambda_E E + \lambda_H H.$$

Completing each scalar square costs at most $\gamma/8$, proving the gap.

In the high-value case choose a finite strategy scoring above $r + \gamma/2$; it exists by the definition of the supremum. Purify its state if necessary, keeping local spaces finite. Its honest Bell realization has $E = H = 0$ and $\langle z, Fz \rangle < -\gamma/2$. To make the Choi baselines faithful, let their identity marginals be fixed and replace

$$R_0 \mapsto R_\varepsilon = (1 - \varepsilon)R_0 + \varepsilon(I_h/h) \otimes I, \quad S_0 \mapsto S_\varepsilon = (1 - \varepsilon)S_0 + \varepsilon(I_h/h) \otimes I.$$

Each physical raw Pauli or Pauli-payload observable is traceless. Thus every centered letter becomes exactly $T_\varepsilon = (1 - \varepsilon)T_0$, while D stays fixed. Expand $F = \sum_w c_w T_w$ and compute a rational $B_{\text{deg}} \geq \sum_w |w| |c_w|$. Telescoping products of contractions gives $\|F(T_\varepsilon) - F(T_0)\| \leq \varepsilon B_{\text{deg}}$. A rational

$$0 < \varepsilon < \min\{1, \gamma/(4 \max\{1, B_{\text{deg}}\})\}$$

preserves a margin below $-\gamma/4$ with faithful baselines. This choice does not require knowing the witness dimension. We do not claim faithful exact-zero anchor witnesses. $\square$

Substitute (16) into F . The result is a rational Hermitian polynomial p in the actual effects Z_A from (15), with the same positive gap and negative witness. This substitution is finite and purely symbolic.

7 Compiling ordered words to inverse Jordan kernels

We give the compiler for scalar polynomials, which is sufficient here. A matrix-coefficient version follows by adding an untouched coefficient factor. Suppose $\alpha I \leq A_j \leq (1 - \alpha)I$, and let

$$p(Z) = \sum_w c_w Z_w, \quad Z_{i_1 \dots i_k} = Z_{i_1} \dots Z_{i_k}, \quad c_{\text{rev}(w)} = \overline{c_w}.$$

Given a positive rational gap g , we construct fixed rational Hermitian residues H_0, H_j^+, H_j^- such that

$$K(R, S) = H_0 \otimes D^{-1} + \sum_j (H_j^+ \otimes B_{A_j}^{-1} + H_j^- \otimes B_{I-A_j}^{-1}) \quad (45)$$

has a uniform gap whenever $p(Z) \geq gI$, and has a negative witness whenever p does.

7.1 An exact edge penalty

For $B = B_{A_j}$ and $u = D^{1/2}x$, $w = D^{1/2}y$, define

$$Q_j(u, w) = \langle u, B^{-1}u \rangle + \langle w - u, (D - B)^{-1}(w - u) \rangle - \langle w, D^{-1}w \rangle.$$

Completing the square in the single Hermitian operator $Z = D^{-1/2}BD^{-1/2}$ gives

$$Q_j(u, w) = \langle x - Zy, [Z(I - Z)]^{-1}(x - Zy) \rangle, \quad 4\|x - Zy\|^2 \leq Q_j(u, w). \quad (46)$$

The identity uses $Z^{-1} + (I - Z)^{-1} = [Z(I - Z)]^{-1}$ and involves no commutation of B with D . Its zero set is exactly the desired edge recurrence $x = Zy$.

7.2 Suffix nodes and explicit residues

Let $\mathcal{W}$ be the smallest suffix-closed set containing the empty word $\emptyset$ and every supported word of p . Write $N_w = |\mathcal{W}|$ and $L_w = \sum_{v \neq \emptyset} |v|$. Each nonempty node $v = jw$ has parent w and the intended recurrence $z_v = Z_j z_w$. Suffix closure preserves the declared noncommutative word order. Define a Hermitian node matrix H_p by

$$(H_p)_{\emptyset, \emptyset} = c_{\emptyset}, \quad (H_p)_{\emptyset, w} = c_w/2, \quad (H_p)_{w, \emptyset} = \overline{c_w}/2 \quad (w \neq \emptyset).$$

Its quadratic form is

$$P(z) = c_{\emptyset} \|z_{\emptyset}\|^2 + \operatorname{Re} \sum_{w \neq \emptyset} c_w \langle z_{\emptyset}, z_w \rangle.$$

On the exact graph it equals $\langle z_{\emptyset}, p(Z) z_{\emptyset} \rangle$. For standard node vectors e_v , and penalty $M_1 \geq 0$, set

$$\begin{aligned} H_0 &= H_p - M_1 \sum_{v=jw} e_w e_w^*, \\ H_j^+ &= M_1 \sum_{v=jw} e_v e_v^*, \quad H_j^- = M_1 \sum_{v=jw} (e_w - e_v)(e_w - e_v)^*. \end{aligned} \quad (47)$$

With $U_v = D^{1/2} z_v$ the kernel form is exactly

$$\langle U, KU \rangle = P(z) + M_1 \sum_{v=jw} Q_j(U_v, U_w). \quad (48)$$

All output residues are constant. No square root of a baseline is required in the algorithm; square roots occur only in this proof.

Lemma 7.1 (Uniform compiler gap). *Choose rational $c'_w \geq |c_w|$, put $b = \sum_{w \neq \emptyset} |w| c'_w$, and set*

$$M_1 = b^2/(8g) + 1/4, \quad \eta = \frac{g}{4N_w + 2gL_w}.$$

If $p(Z) \geq gI$ on the actual domain, then

$$K(R, S) \geq \eta I \otimes D^{-1} \quad (49)$$

on that domain. Any strict negative witness for p transfers exactly to K at the same baselines and reference sizes.

Proof. For arbitrary nodes let $a = \|z_{\emptyset}\|$, $r_{jw} = z_{jw} - Z_j z_w$, and $t^2 = \sum_{v \neq \emptyset} \|r_v\|^2$. Expanding the recurrence along each path, using contractions, gives

$$\|z_v - Z_v z_{\emptyset}\| \leq \sqrt{|v|} t, \quad \sum_v \|z_v - Z_v z_{\emptyset}\|^2 \leq L_w t^2.$$

Since the objective has only root-to-word terms,

$$|P(z) - \langle z_{\emptyset}, p(Z) z_{\emptyset} \rangle| \leq bat.$$

By (46) and Young's inequality,

$$\langle U, KU \rangle \geq ga^2 - bat + 4M_1 t^2 \geq (g/2)a^2 + t^2.$$

Also $\|z\|^2 \leq 2N_w a^2 + 2L_w t^2$, which proves (49) with the displayed conservative η , including the constant-polynomial case. If $\langle \xi, p(Z) \xi \rangle < 0$, set $z_v = Z_v \xi$ and $U_v = D^{1/2} z_v$. Every edge penalty is zero, so (48) equals the negative polynomial value exactly. $\square$

All constants and residues are rational complex when the polynomial and effects are. Bounds such as $c'_w = |\operatorname{Re} c_w| + |\operatorname{Im} c_w|$ suffice. There are at most $2g_0 + 1$ distinct denominators for g_0 input letters, each a fixed strictly positive physical effect, and $N_w \leq 1 + \sum_{w \neq \emptyset, c_w \neq 0} |w|$. The positive gap is essential for this finite-penalty argument. Mere nonnegativity of p is not being compiled by the assertion.

8 Realizing complex residues by positive measurement effects

We now start with a finite rational Hermitian kernel

$$K(R, S) = \sum_r H_r \otimes B_{A_r}(R, S)^{-1}, \quad A_r > 0, \quad (50)$$

and a supplied gap $K \geq \eta I \otimes B_I^{-1}$, or a strict negative witness. Let the residue dimension be d and the denominator physical dimension be p_0 . Choose q_0 with $p_0 q_0 \geq d$ and distinct physical matrix units $P_\alpha = |a_\alpha\rangle\langle p_0 + b_\alpha|$ in the upper-to-lower block. For $v \in \mathbb{C}^d$ set $V(v) = \sum_\alpha v_\alpha P_\alpha$.

8.1 Rational signed rank one decomposition

Every rational Hermitian H_r has a finite decomposition

$$H_r = \sum_t c_t v_t v_t^*, \quad c_t \in \mathbb{Q} \setminus \{0\}, \quad v_t \in \mathbb{Q}(i)^d. \quad (51)$$

One elementary construction uses diagonal units and the vectors $e_j \pm e_k$, $e_j \pm ie_k$: differences of their outer products span the real and imaginary off-diagonal Hermitian matrix units. Thus no irrational spectral square roots are needed.

For every signed term, make *two* baseline outcomes with upper block $\lambda A_r / |c_t|$ and lower block $\lambda \text{Tr}(A_r) / (p_0 |c_t|) I_{q_0}$. Choose a small positive rational λ so that the sum of all upper blocks is strictly less than I_{p_0} , and add a common filler

$$D_0 = \text{diag}(A_0, [\text{Tr } A_0 / p_0] I_{q_0}), \quad A_0 = I_{p_0} - \sum_{t, \ell=1,2} \lambda A_{r(t)} / |c_t| > 0.$$

The baseline list (D_i) is a strictly positive rational POVM. For term t its two Hermitian off-diagonal numerators are

$$O_{t,1} = V(v_t) + V(v_t)^*, \quad O_{t,2} = iV(v_t) - iV(v_t)^*.$$

Put these numerators on the source side if $c_t > 0$, on the target side if $c_t < 0$, and put zero on the other side. The filler has zero numerator on both sides. Denote the resulting lists by O_i^E, O_i^F .

8.2 Effective baselines and both quadratures

For a faithful physically block-diagonal base $\bar{R} = \text{diag}(R_0, T)$, let

$$R_{\text{eff}} = R_0 + (I_{p_0} / p_0) \otimes \text{Tr}_{q_0} T, \quad W = \text{Tr}_{p_0} R_{\text{eff}} = \text{Tr}_{p_0+q_0} \bar{R}. \quad (52)$$

The output of D_i is exactly the corresponding scaled $\mathcal{A}_{A_r}(R_{\text{eff}})$. Conversely every $R_{\text{eff}} > 0$ is realizable with both blocks faithful by taking $T = tI$ for sufficiently small $t > 0$ and subtracting its contribution from R_{eff} .

Write a physical cross tangent as

$$X = \sum_{\alpha=1}^d (P_\alpha \otimes Z_\alpha + P_\alpha^* \otimes Z_\alpha^*).$$

For one term let $U_t = \sum_\alpha \overline{(v_t)_\alpha} Z_\alpha$. Its two outputs are $U_t + U_t^*$ and $iU_t^* - iU_t$, up to an immaterial overall sign for the second. Because Ω^{-1} respects adjoints, their squared norms

sum to $4\langle U_t, \Omega^{-1}(U_t) \rangle$. In particular the two quadratures eliminate extra conjugate cross terms for arbitrary tangents, not just for one-sided rectangular slices. Define

$$\Gamma_{\bar{R}}(X) = \sum_i \left[g_{\bar{R}_i}(\text{Tr}_H(O_i^E X)) - g_{\bar{R}_i}(\text{Tr}_H(O_i^F X)) \right], \quad \bar{R}_i = \text{Tr}_H(D_i \bar{R}). \quad (53)$$

The precise normalization is

$$\Gamma_{\bar{R}}(X) = \frac{8}{\lambda} \langle \text{vec } Z, K(R_{\text{eff}}, R_{\text{eff}}) \text{vec } Z \rangle. \quad (54)$$

The factor eight is 4 from the quadratures and 2 from $\Omega^{-1} = 2\mathcal{J}^{-1}$; the coefficient $|c_t|/\lambda$ is absorbed by the baseline scaling.

Let $\mathcal{C}$ be the real span of the off-diagonal numerator matrices. Give its coefficients the Hilbert–Schmidt orthonormal physical-coordinate norm

$$|X|_W^2 = \sum_{\mu} g_W(X_{\mu}), \quad X = \sum_{\mu} C_{\mu} \otimes X_{\mu}.$$

On the full selected complex-coordinate space,

$$|X|_W^2 = 4 \sum_{\alpha} \langle Z_{\alpha}, \mathcal{J}_{W,W}^{-1}(Z_{\alpha}) \rangle. \quad (55)$$

The same identity restricts to $\mathcal{C}$. Any physical cross direction orthogonal to $\mathcal{C}$ is invisible to both signal lists and will require no coercivity. From (54) and the kernel gap we get the exact cross gap

$$\Gamma_{\bar{R}}(X) \geq \gamma_{\text{cross}} |X|_W^2, \quad \boxed{\gamma_{\text{cross}} = 2\eta/\lambda}. \quad (56)$$

For a negative rectangular kernel witness at R, S , use effective baseline $R \oplus S$ on a direct-sum reference and put each Z_{α} only in its R -to- S reference block. The square-matrix form restricts exactly to the rectangular Sylvester form. Formula (54) is strictly negative. Realize $R \oplus S$ by faithful physical lower padding using (52). If necessary project the tangent onto $\mathcal{C}$; its signal outputs and negative value are unchanged. This constructs a faithful physical-block-diagonal negative slice witness.

9 A global completion valid for every state and tangent

The slice construction alone does not prove a full channel order. Coherent physical baselines change the denominators, and physical diagonal tangent directions remain. We now control both uniformly.

9.1 Signal pairs and informationally complete padding

Given (D_i) and O_i^E, O_i^F above, define signal POVMs

$$E_{i,\pm} = (D_i \pm \delta O_i^E)/2, \quad F_{i,\pm} = (D_i \pm \delta O_i^F)/2.$$

Their sums are I . For sufficiently small rational $\delta > 0$, every effect is strictly positive. Choose a strictly positive rational block-diagonal informationally complete POVM (G_h) whose real span is the entire physical block-diagonal Hermitian subspace. For example, for a rational Hermitian basis (B_{μ}) choose sufficiently small rational $t > 0$ and use effects $(I \pm tB_{\mu})/(2L)$, where L is the number of basis elements. This gives exact normalization and spanning.

If $(\tilde{B}_\mu)$ is an orthonormal basis of that subspace, let

$$g_{h\mu} = \text{Tr}(G_h \tilde{B}_\mu), \quad c > 0 \quad \text{be a certified lower bound for } \lambda_{\min}\left(\sum_h g_h g_h^\top\right).$$

Such a positive rational lower bound is effectively computable by finite algebraic estimates. The basis used for estimates need not be rational; all physical effects are rational. The final source POVM consists of $E_{i,\pm}/2$ and $G_h/2$. The final target consists of $F_{i,\pm}/2$ and a single erasure effect $I/2$. They are normalized strictly positive rational POVMs once δ is chosen.

9.2 A reference score gauge

For any faithful physical-reference base R , let $W = \text{Tr}_H R$. For a Hermitian tangent X , put

$$A = \Omega_W^{-1}(\text{Tr}_H X), \quad X' = X - \Omega_R(I_H \otimes A).$$

Then $\text{Tr}_H X' = 0$. Every physical trace-preserving channel Φ obeys $\Phi_R(\Omega_R(I \otimes A)) = \Omega_{\Phi_R(R)}(I \otimes A)$. Expansion of the inverse-metric square shows

$$g_{\Phi_R(R)}(\Phi_R(X')) - g_{\Phi_R(R)}(\Phi_R(X)) = -2 \text{Tr}[X(I \otimes A)] + \text{Tr}[R(I \otimes A^2)],$$

independently of Φ . Every source-minus-target difference is therefore unchanged by this gauge.

Decompose $X' = Y + X_C + X_{\text{inv}}$ into physical diagonal, relevant cross, and invisible cross parts. The erasure branch sees zero. The source padding sees only Y and contributes at least

$$\sum_h g_{\text{Tr}_H(G_h R)}(\text{Tr}_H(G_h Y)) \geq c \|Y\|_W^2. \quad (57)$$

Indeed $G_h \leq I$ makes its base at most W , so inverse Jordan ordering bounds each term below by its W -norm; the frame lower bound gives the last inequality. The gauge is essential: no false coercivity assertion is made on reference-only tangents before this quotient.

9.3 An exact paired resolvent identity

For one signal outcome on either side abbreviate

$$\begin{aligned} R_i &= \text{Tr}_H(D_i R), \quad V = \text{Tr}_H(OR), \quad U = \text{Tr}_H(D_i Y), \quad Z = \text{Tr}_H(OX_C), \\ J &= \Omega_{R_i}, \quad K = \Omega_V, \quad u = J^{-1/2}U, \quad z = J^{-1/2}Z, \quad T = J^{-1/2}KJ^{-1/2}. \end{aligned}$$

The sum of the two sign-outcome Fisher costs is exactly

$$F_{\text{pair}} = \|u\|^2 + \delta^2 \langle z - Tu, (I - \delta^2 T^2)^{-1} (z - Tu) \rangle. \quad (58)$$

To verify it, write the pair as one half the sum of the inverse quadratic forms of $(u \pm \delta z, I \pm \delta T)$. The even inverse average is $(I - \delta^2 T^2)^{-1}$ and the odd difference is $-\delta T(I - \delta^2 T^2)^{-1}$. Expanding proves the completed square. Every odd order cancels. The base term $\|u\|^2$ is identical on source and target and cancels in their difference.

9.4 Dimension independent error bounds

Choose rational lower bounds $D_i \geq d_i I$, $d_i > 0$, and certified rational upper bounds in the following definitions:

$$\begin{aligned}\kappa &\geq \max_{i,s=E,F} \|O_i^s\|_{\text{op}}/d_i, \\ B_z &\geq \sum_i d_i^{-1} (\|O_i^E\|_{\text{HS}}^2 + \|O_i^F\|_{\text{HS}}^2), \\ B_u &\geq 2 \sum_i d_i^{-1} \|D_i\|_{\text{HS}}^2.\end{aligned}\tag{59}$$

We may take all positive; a zero signal case is trivial. Since $R_i \geq d_i W$ and $-\|O\|_{\text{op}} W \leq V \leq \|O\|_{\text{op}} W$, one has $\|T\| \leq \kappa$ uniformly. Coefficient-frame Cauchy–Schwarz and inverse Jordan ordering imply

$$\sum_{i,s} \|z_i^s\|^2 \leq B_z |X_C|_W^2, \quad \sum_{i,s} \|u_i\|^2 \leq B_u |Y|_W^2.$$

For $\delta\kappa \leq 1/2$,

$$0 \leq (I - \delta^2 T^2)^{-1} - I \leq \frac{4}{3} \delta^2 \kappa^2 I.$$

Let $a = |X_C|_W$, $b = |Y|_W$, and $\gamma = \gamma_{\text{cross}}$. The gap (56) applies to the pinching of R ; its R_i and W are exactly unchanged by that pinching. Thus the leading z -square source-minus-target difference is at least γa^2 . Expanding (58), bounding the mixed terms by Cauchy–Schwarz, and using the last resolvent bound gives

$$\begin{aligned}\delta^{-2} \Delta_{\text{signal}} &\geq \gamma a^2 - 2\kappa \sqrt{B_z B_u} ab - \kappa^2 B_u b^2 \\ &\quad - \frac{8}{3} \delta^2 \kappa^2 (B_z a^2 + \kappa^2 B_u b^2).\end{aligned}\tag{60}$$

This includes all physical coherences through V and T . If

$$\delta^2 \leq \frac{3\gamma}{16\kappa^2 B_z}, \quad \delta\kappa \leq \frac{1}{2},\tag{61}$$

then the resolvent cross loss is at most $\gamma a^2/2$. Young’s inequality absorbs the mixed term using another $\gamma a^2/4$ and $4\kappa^2 B_z B_u b^2/\gamma$. The diagonal resolvent loss is at most $(2/3)\kappa^2 B_u b^2$. Hence

$$\Delta_{\text{signal}} \geq \delta^2 \left[\frac{\gamma}{4} a^2 - C_0 b^2 \right], \quad C_0 = \frac{5}{3} \kappa^2 B_u + \frac{4\kappa^2 B_z B_u}{\gamma}.\tag{62}$$

The final POVMs give equal weight $1/2$ to signal and padding. Equations (57) and (62) prove nonnegative total Fisher difference whenever, in addition,

$$\delta^2 \leq c/C_0.\tag{63}$$

Choose a positive dyadic rational δ satisfying (61)–(63). The same condition $\delta\kappa \leq 1/2$ ensures $D_i \pm \delta O_i^s \geq (d_i/2)I$, so all final effects are strictly positive.

Proposition 9.1 (Global completion). *The preceding finite rational construction converts the supplied kernel gap into complete SLD dominance of the final source POVM over the target POVM. A negative kernel witness converts into a genuine finite faithful SLD violation.*

Proof. The positive assertion was proved for every faithful base and every Hermitian tangent after an invariant gauge, so it is complete dominance. For the negative assertion use the

physically block-diagonal base and pure cross tangent of Section 8. Then $V = 0$ and $U = 0$ in every pair, and both padding tangents vanish. The final difference is exactly

$$\frac{\delta^2}{2}\Gamma_R(X) < 0.$$

It is an exact metric witness with no perturbative error. If the base is unnormalized, divide both it and the tangent by its trace; the difference is divided by the same positive scalar. The tangent is traceless, and the base is faithful. $\square$

10 Computability classification and the reference obstruction

Apply the game reduction with $r = 3/4$ and $\gamma = 1/4$. In the nonhalting case the external game has $\omega^* \leq 1/2 = r - \gamma$, so Proposition 6.1 gives a uniform polynomial gap $g = \gamma/2 = 1/8$. Lemma 7.1 yields a positive inverse-kernel gap, and Proposition 9.1 yields complete SLD dominance.

In the halting case $\omega^* = 1$, so a finite strategy scoring above $7/8 = r + \gamma/2$ exists, whether or not value one is attained. The faithful polynomial witness transfers exactly through the graph compiler and the negative POVM slice. Complete SLD dominance fails. Every output operation uses finite rational arithmetic and effective bounds; no witness dimension is needed to compute the POVMs. We have therefore reduced halting to failure and nonhalting to dominance.

10.1 Failure is recursively enumerable

Enumerate reference dimensions and rational complex faithful density matrices ρ together with rational complex traceless Hermitian tangents H . Exact rational linear algebra decides Hermiticity, normalization and strict positive definiteness. For each strictly positive rational POVM and faithful rational input the output blocks are positive definite and rational. Solve each rational Sylvester system

$$\Omega_{\rho_i^E}(L_i) = H_i^E$$

exactly. Its Fisher value is rational, so the source-minus-target sign is decidable. Every strict real violation persists on an open neighborhood by continuity of inversion on positive definite matrices. Rational faithful density matrices and rational traceless Hermitian tangents are dense in the required affine spaces. Thus the enumeration eventually detects every failure. It may run forever precisely on dominating pairs.

Failure is in RE and is RE-hard by the halting reduction; hence it is RE-complete. The dominating set is coRE-complete. By Section 2, exactly the same set of pairs defines the unrestricted entropy order, proving assertions 1 and 2 of Theorem 1.2.

10.2 No computable sufficient SLD reference cutoff

For a fixed reference dimension n , existence of a faithful SLD violation is a first-order real-algebraic question. Use real and imaginary matrix coordinates for ρ, H, L_i ; impose trace and Hermiticity constraints, positive definiteness by principal-minor conditions, and the polynomial Sylvester equations. The Fisher difference is polynomial in these variables once the scores L_i are included. Real quantifier elimination therefore decides whether a violation exists at that fixed size.

Suppose a total computable bound $b(E, F)$ were sufficient to detect every failure. Compute it and decide the finitely many reference sizes $1, \dots, b(E, F)$ by quantifier elimination. Absence of a violation at those sizes would certify complete dominance, while its presence would certify failure. This contradicts the undecidability just proved. Assertion 3 follows.

This argument specifically establishes the reference-cutoff obstruction for SLD tests. The entropy decision problem is equally undecidable by order equivalence; no fixed-dimension algorithm for transcendental entropy formulas is assumed in the cutoff argument.

11 A bounded entropy defect

For strictly positive POVMs E, F define

$$\mathcal{V}(E, F) = \sup_{R, \rho, \sigma} [D(\mathcal{F}_R(\rho) \| \mathcal{F}_R(\sigma)) - D(\mathcal{E}_R(\rho) \| \mathcal{E}_R(\sigma))], \quad (64)$$

where R ranges over finite untouched references and ρ, σ are faithful input states. In particular every difference in this definition is between finite numbers.

Lemma 11.1 (Boundedness of the defect). *If $a_j I \leq F_j \leq b_j I$ with $a_j > 0$, then*

$$0 \leq \mathcal{V}(E, F) \leq \max_j \log(b_j/a_j) < \infty.$$

Proof. For any strictly positive POVM $M = (M_i)$, choose $\alpha_i I \leq M_i \leq \beta_i I$ with $\alpha_i > 0$. Let ρ_R, σ_R be the reference marginals and ρ_i, σ_i the measurement output blocks. Positivity of partial trace gives $\alpha_i \rho_R \leq \rho_i \leq \beta_i \rho_R$, and similarly for σ_i . Operator monotonicity of logarithm implies

$$(\log \alpha_i)I \leq \log \rho_i - \log \rho_R \leq (\log \beta_i)I,$$

with the same bounds for $\log \sigma_i - \log \sigma_R$. Since $\sum_i \rho_i = \rho_R$, the reference contribution cancels exactly in

$$\begin{aligned} G_M(\rho, \sigma) &:= D(\mathcal{M}_R(\rho) \| \mathcal{M}_R(\sigma)) - D(\rho_R \| \sigma_R) \\ &= \sum_i \text{Tr } \rho_i [(\log \rho_i - \log \rho_R) - (\log \sigma_i - \log \sigma_R)]. \end{aligned}$$

Data processing under forgetting the outcome gives $G_M \geq 0$; the displayed logarithm bounds give $G_M \leq \max_i \log(\beta_i/\alpha_i)$. The defect in (64) equals $G_F - G_E$, yielding the upper bound. Taking $\rho = \sigma$ proves nonnegativity of the supremum. $\square$

Boundedness alone does not provide a computable approximation procedure. To rule one out we need more than the strict witnesses used in the exact-order reduction: their absolute entropy margin must have a positive lower bound computable from the game, without knowing the strategy dimension. The following two sections establish this extra normalization.

12 Maximally entangled completeness and normalized margins

We use a more precise part of the same external theorem. Theorem 12.6 of [1], printed page 172 of arXiv:2001.04383v3, supplies a finite-dimensional perfect *PCC* strategy in the halting case, while retaining soundness at most 1/2 in the nonhalting case. Definitions 5.9–5.11 on printed pages 51–52 specify projective measurements whose every answer projector M , in both players' families, satisfies

$$(M \otimes I)\psi = (I \otimes M)\psi. \quad (65)$$

The additional question-pair commutation condition in PCC is not needed below.

Lemma 12.1 (A perfect maximally entangled sector). *A finite perfect strategy with the consistency property (65) has a finite perfect maximally entangled strategy obtained by restriction to a Schmidt sector.*

Proof. Write $\psi = \text{vec } C$, with $\|C\|_{\text{HS}} = 1$. Consistency says $MC = CM^\top$. Since M is Hermitian, its adjoint relation gives

$$MCC^* = CM^\top C^* = CC^*M.$$

Thus every Alice answer projector preserves the spectral sectors of her reduced state CC^* . The same reasoning for the other family proves that every Bob answer projector preserves the matching sectors of $(C^*C)^\top$. For each positive Schmidt eigenvalue λ of multiplicity d_λ , the state decomposes as

$$\psi = \sum_\lambda \sqrt{\lambda d_\lambda} \psi_\lambda, \quad \sum_\lambda \lambda d_\lambda = 1,$$

where ψ_λ is a unit maximally entangled vector between the matching d_λ -dimensional sectors. Sector preservation and orthogonality make every game probability the corresponding convex combination of sector probabilities. Since payoffs lie in $[0, 1]$, a score of one forces each nonzero sector to score one. Restrict both measurement families to any such sector. $\square$

This is a property of the hard game family. No maximally entangled optimality assertion for arbitrary games is used.

12.1 Faithful Choi regularization with a fixed marginal

Keep $r = 3/4$, $\gamma = 1/4$ in the reduction. Let $a = 2MK$ be the initial physical dimension and let $L = N_w$ be the compiler node count. Use lower physical dimension $q_0 = \lceil L/a \rceil$ in Section 8. Write $\lambda > 0$ for its baseline scaling and $\delta_s > 0$ for the signal parameter of Section 9. All are computed from the game.

Choose rational $B_{\text{deg}} \geq \sum_w |w| |c_w|$ for the centered-letter polynomial $F = \sum_w c_w T_w$, and put

$$e = \min\{1/2, \gamma/(8 \max\{1, B_{\text{deg}}\})\}. \quad (66)$$

A perfect maximally entangled strategy, together with the Bell anchor, has local dimension ℓ and root $z = \text{vec}(I_\ell)/\sqrt{\ell}$ after choosing local bases. Its Choi baselines have reference marginal I_ℓ . Regularize them by

$$R_e = (1 - e)R_0 + e(I_a/a) \otimes I_\ell, \quad S_e = (1 - e)S_0 + e(I_a/a) \otimes I_\ell.$$

Their marginals stay fixed and

$$(e/a)I \leq R_e, S_e \leq aI. \quad (67)$$

For the upper bound, any positive block matrix with physical partial trace I obeys $R \leq a \sum_i |i\rangle\langle i| \otimes R_{ii} \leq aI$ by the pinching inequality and $\sum_i R_{ii} = I$. The perfect honest polynomial expectation is $r - 1 = -\gamma$. Every centered letter is multiplied by $1 - e$, so

$$\langle z, F(T_e)z \rangle \leq -\gamma + eB_{\text{deg}} \leq -\gamma/2. \quad (68)$$

12.2 Operator norm control of word witnesses

At these marginal-identity baselines, $D = 2I$. Upon reshaping, each normalized positive-effect letter acts as

$$Z \mapsto (C_A Z + Z D_A)/2, \quad 0 \leq C_A, D_A \leq I,$$

up to the fixed vectorization transpose. It therefore contracts operator norm. Starting from $I_\ell/\sqrt{\ell}$, every word-node matrix has operator norm at most $1/\sqrt{\ell}$. The exact graph coordinates $U_w = \sqrt{2}z_w$ have reshaped norms at most $\sqrt{2}/\sqrt{\ell}$.

On the reference $\mathbb{C}^\ell \oplus \mathbb{C}^\ell$, put these matrices in the upper-right reference block of the L selected physical cross coordinates. Let Z be the resulting rectangular physical block and $X = \begin{pmatrix} 0 & Z \\ Z^* & 0 \end{pmatrix}$ the Hermitian physical-cross tangent. Triangle inequality gives

$$\text{Tr } X = 0, \quad \|X\| \leq 2L/\sqrt{\ell}. \quad (69)$$

The operator norm, rather than only a Hilbert–Schmidt bound, is the reason maximally entangled completeness is useful here.

12.3 Physical padding and a dimension independent Fisher gap

Let $R_{\text{eff}} = R_e \oplus S_e$, and define the physical-block-diagonal baseline

$$T_{\text{low}} = \frac{e}{2q_0} I_{q_0} \otimes I_{2\ell}, \quad R_{\text{top}} = R_{\text{eff}} - \frac{e}{2a} I_a \otimes I_{2\ell}, \quad \Theta_0 = \text{diag}(R_{\text{top}}, T_{\text{low}}).$$

Its effective baseline (52) is exactly R_{eff} , its reference marginal is $I_{2\ell}$, and $\text{Tr } \Theta_0 = 2\ell$. Moreover

$$\frac{e}{2 \max\{a, q_0\}} I \leq \Theta_0 \leq aI.$$

Normalize by

$$\rho = \Theta_0/(2\ell), \quad H = X/\sqrt{\ell}, \quad h = \frac{8L \max\{a, q_0\}}{e}. \quad (70)$$

Equations (69)–(70) imply

$$\frac{e}{4 \max\{a, q_0\} \ell} I \leq \rho \leq \frac{a}{2\ell} I, \quad \|H\| \leq \frac{2L}{\ell}, \quad -h\rho \leq H \leq h\rho.$$

Let $\alpha > 0$ be a certified rational lower bound for every final effect on both sides. For either final channel Φ , set $A_\Phi = \Phi_R(\rho)$ and $K_\Phi = \Phi_R(H)$. Since $\rho_R = I_{2\ell}/(2\ell)$, every outcome block obeys

$$\frac{\alpha}{2\ell} I \leq (A_\Phi)_j \leq \frac{1}{2\ell} I.$$

Thus each full output state has condition number at most $\kappa = 1/\alpha$, independently of ℓ . Positivity also preserves $-hA_\Phi \leq K_\Phi \leq hA_\Phi$.

For the unnormalized base and cross tangent, the exact graph identity and (54) give

$$\Gamma_{\Theta_0}(X) = \frac{8}{\lambda} \langle z, F(T_e)z \rangle.$$

The final signal/padding branch weight is $1/2$ and its signal factor is δ_s^2 . Hence the source-minus-target Fisher difference is at most $-2\delta_s^2\gamma/\lambda$ on (Θ_0, X) . The change to (ρ, H) multiplies this difference by 2, because the metric is inverse-linear in its base and quadratic in its tangent. Consequently

$$g_{A_{\mathcal{F}}}(K_{\mathcal{F}}) - g_{A_{\mathcal{E}}}(K_{\mathcal{E}}) \geq \delta_G, \quad \delta_G = 4\delta_s^2\gamma/\lambda > 0. \quad (71)$$

All three constants h, κ, δ_G are positive rational bounds computable solely from the game description. Neither the faithful normalization nor the absolute Fisher gap loses an unknown factor of the strategy dimension.

13 A uniform conversion from Fisher gap to entropy gap

The following quantitative refinement of Section 2 supplies the needed positive entropy margin.

Lemma 13.1 (Uniform entropy witness). *Suppose a faithful input state ρ and Hermitian H satisfy $-h\rho \leq H \leq h\rho$ for rational $h > 0$. Suppose both effective output states of channels $\mathcal{P}, \mathcal{Q}$ have condition number at most rational $\kappa \geq 1$, and their target-minus-source SLD gap is at least rational $d_0 > 0$. There is a computable rational $\varepsilon_0 > 0$ and a finite auxiliary reference factor depending only on h, κ, d_0 , for which faithful input state pairs attain target-minus-source Umegaki gap at least ε_0 .*

Proof. Choose the integer

$$m = \max\{6, \lceil 288\kappa^2 h^4 / d_0^2 \rceil\}, \quad N = 2m + 1, \quad r_0 = N + 1. \quad (72)$$

Use the poles $\lambda_j = 1 + j/m^3$ and partial-fraction coefficients c_j of Lemma 2.1. Define positive rational constants

$$\begin{aligned} \Lambda_0 &= \sum_{j=0}^{2m} |c_j|, & q_* &= h\kappa\Lambda_0, & j_* &= 5^{-(4m+2)}/N, \\ \eta_0 &= \min\{1/4, j_*d_0/(16q_*^2)\}, \\ t_0 &= \min\{1/(2q_*), 3j_*d_0/(256q_*^3)\}, & \varepsilon_0 &= j_*d_0t_0^2/16. \end{aligned} \quad (73)$$

On an auxiliary centre and N leaves take

$$W = \text{diag}(1, \lambda_0, \dots, \lambda_{2m}), \quad v = N^{-1/2} \sum_j e_{j+1}, \quad w = \sum_j c_j e_{j+1},$$

$$\tau = vv^*, \quad L = e_0 w^* + w e_0^*, \quad \sigma_{\eta_0} = (1 - \eta_0)\tau + \eta_0 I_{r_0}/r_0, \quad Z_0 = \text{Tr } W.$$

The matrices $W, L, \tau, \sigma_{\eta_0}$ are rational even though v contains a square root, and $\|L\| \leq \Lambda_0$. Set

$$\alpha = \rho \otimes \sigma_{\eta_0}, \quad \beta(t) = (\rho \otimes W + tH \otimes L)/Z_0. \quad (74)$$

Both have trace one, since $\text{Tr } L = 0$. The numerator is faithful. The relative tangent bound gives

$$\|(\rho \otimes W)^{-1/2}(H \otimes L)(\rho \otimes W)^{-1/2}\| \leq h\Lambda_0 \leq q_*.$$

Thus $\beta(t) > 0$ and its raw denominator is at least $(\rho \otimes W)/2$ when $|t| \leq 1/(2q_*)$.

Uniform filter error. The implemented filter is $F(s) = F_m(s)/\sqrt{N}$, with mass $J_m = \int_0^\infty |F(s)|^2 ds = I_m/N$. For the beta-prime density proportional to f_m^2 , direct integration gives

$$\mathbb{E}(T - 1)^2 = \frac{2(m+1)}{m(2m-1)} \leq \frac{2}{m} \quad (m \geq 2).$$

The multiplicative filter bounds in Lemma 2.1, including normalization, therefore imply for $\nu_m(ds) = F_m(s)^2 ds/I_m$ that

$$\mathbb{E}_{\nu_m}|s - 1| \leq e^{4/m+2/m^2} \sqrt{2/m} \leq 3\sqrt{2/m} \quad (m \geq 6). \quad (75)$$

For eigenvalues a, b of either one output state,

$$\frac{|(b + as)^{-1} - (a + b)^{-1}|}{(a + b)^{-1}} = \frac{a|s - 1|}{b + as} \leq \kappa|s - 1|.$$

Also if $K = A^{1/2}TA^{1/2}$ with $\|T\| \leq h$, then

$$g_A(K) \leq \text{Tr}(KA^{-1}K) = \text{Tr}(AT^2) \leq h^2. \quad (76)$$

The first inequality follows from $2/(a+b) \leq (1/a + 1/b)/2$ and Hermiticity of K . Let h_Φ be the pure-leaf-numerator entropy Hessian for channel Φ . The exact formula (11) and (75)–(76) give

$$|h_\Phi/J_m - g_{A_\Phi}(K_\Phi)| \leq 3\kappa h^2 \sqrt{2/m}.$$

By (72), the two-channel error is at most $d_0/2$. Hence $h_Q - h_P \geq J_m d_0/2$. For $s \in [1, 2]$ all $\lambda_j \leq 3$, so $F_m(s)^2 \geq 5^{-(4m+2)}$ and $J_m \geq j_*$. No numerical integration of the large partial-fraction coefficients is needed.

Dimension independent logarithm derivatives. For either output and any auxiliary density matrix σ , the variable part of its entropy is

$$-\text{Tr}[(A \otimes \sigma) \log(B + tY)], \quad B = A \otimes W, \quad Y = K \otimes L.$$

If $\mu = \lambda_{\min}(A)$, then $\|Y\| \leq h\lambda_{\max}(A)\Lambda_0 \leq q_*\mu$. For $|t| \leq 1/(2q_*)$, $B + tY \geq \mu I/2$. Resolvent differentiation gives

$$\frac{d^n}{dt^n} \log(B + tY) = (-1)^{n+1} n! \int_0^\infty R_t(u) [Y R_t(u)]^n du, \quad R_t(u) = (B + tY + uI)^{-1}.$$

Taking norms and integrating yields

$$\left\| \frac{d^n}{dt^n} \log(B + tY) \right\| \leq (n-1)! \left(\frac{\|Y\|}{\lambda_{\min}(B + tY)} \right)^n.$$

Since $\text{Tr}(A \otimes \sigma) = 1$, each channel's absolute second derivative at zero is at most q_*^2 , and its absolute third derivative throughout the interval is at most $16q_*^3$.

Faithful numerator and a finite perturbation. Let $G_\sigma(t)$ be the target-minus-source entropy difference with numerator $\rho \otimes \sigma$ and denominator $\beta(t)$. Saturation gives $G_\sigma(0) = 0$. For $\sigma = \tau, I_{r_0}/r_0, \sigma_{\eta_0}$, conjugating the centre coordinate by -1 leaves the numerator fixed and sends L to $-L$. Thus G_σ is even and $G'_\sigma(0) = 0$. The second denominator derivative is linear in σ . Hence

$$G''_\tau(0) \geq j_* d_0/2, \quad |G''_{I_{r_0}/r_0}(0)| \leq 2q_*^2,$$

and the choice of η_0 implies

$$G''_{\sigma_{\eta_0}}(0) \geq \frac{3}{4}(j_* d_0/2) - 2\eta_0 q_*^2 \geq j_* d_0/4.$$

The two-channel third derivative has absolute value at most $32q_*^3$. Taylor's theorem now gives

$$G_{\sigma_{\eta_0}}(t) \geq (j_* d_0/8)t^2 - (16/3)q_*^3|t|^3.$$

At $t = t_0$, the cubic term is at most $(j_* d_0/16)t_0^2$, so the gap is at least ε_0 . Finally $\sigma_{\eta_0} \leq I \leq W$ and the denominator lower bound imply $\beta(t) \geq \alpha/(2Z_0)$. This inequality survives either channel. Operator monotonicity of logarithm gives each witness divergence in $[0, \log(2Z_0)]$. Thus the construction involves finite faithful divergences and no infinity-minus-infinity expression. $\square$

The auxiliary factor r_0 and all constants in this lemma are independent of the original reference dimension. They may be extremely large or small, respectively. The lemma is an effective margin statement, not an efficiency claim or a bound on the original reference size. Its state pairs need not have equal reference marginals.

14 The additive approximation obstruction

Theorem 14.1 (No total additive approximation algorithm). *There is no total algorithm which, given rational complex strictly positive POVMs E, F and rational $\varepsilon > 0$, returns a rational v satisfying*

$$|v - \mathcal{V}(E, F)| \leq \varepsilon.$$

The conclusion holds with the supremum in (64) restricted to faithful state pairs, for which the value is finite and nonnegative.

Proof. Given a Turing machine, compute its hard game and the POVMs from the preceding reduction. In the nonhalting case complete SLD and therefore complete entropy dominance hold, so $\mathcal{V} = 0$. In the halting case Lemma 12.1 gives a perfect maximally entangled strategy. Section 12 supplies a faithful Fisher witness and game-computable rational constants h, κ, δ_G satisfying Lemma 13.1. Apply that lemma with $d_0 = \delta_G$ to compute a positive rational ε_G and a faithful entropy witness with defect at least ε_G . All constants can be computed without finding the strategy or its dimension. Thus halting implies $\mathcal{V} \geq \varepsilon_G$.

A purported approximation algorithm at tolerance $\varepsilon_G/4$ would return $v \leq \varepsilon_G/4$ in the nonhalting case and $v \geq 3\varepsilon_G/4$ in the halting case. Comparing with $\varepsilon_G/2$ would decide halting, a contradiction. $\square$

The primitive gap ε_G depends effectively on the game. This is enough to amplify to a universal numerical accuracy obstruction.

Corollary 14.2 (Fixed additive error is already undecidable). *There is no total algorithm which approximates $\mathcal{V}(E, F)$ to additive error $1/4$ on every finite rational complex strictly positive POVM pair.*

Proof. First, unrestricted complete entropy dominance tensorizes. If $\mathcal{E} \succeq_{\text{RE}}^c \mathcal{F}$, then for any joint input state pair to k copies, replace the channels $\mathcal{E}$ by $\mathcal{F}$ one factor at a time. At each replacement, all other unprocessed inputs and already-produced outputs are an untouched finite reference for the one-factor comparison. The order applies to every state pair, without a marginal constraint. Chaining the inequalities yields

$$\mathcal{E}^{\otimes k} \succeq_{\text{RE}}^c \mathcal{F}^{\otimes k}.$$

For the hard instance compute the positive rational ε_G and choose $k = \lceil 1/\varepsilon_G \rceil$. The product effects of the tensor-power POVMs are still rational, strictly positive and normalized. In the nonhalting case tensorization gives $\mathcal{V}(E_G^{\otimes k}, F_G^{\otimes k}) = 0$. In the halting case use the k -fold products of the faithful entropy-witness states. Relative entropy is additive on tensor products, so their defect is at least $k\varepsilon_G \geq 1$. An additive- $1/4$ approximation is at most $1/4$ in the first case and at least $3/4$ in the second. Thresholding at $1/2$ decides halting, which is impossible. $\square$

The tensor-power output can be enormous; this is a computable reduction, not a polynomial-time claim. Exact-order undecidability alone would not establish the fixed-error result. The computable primitive margin, its dimension-independent normalization, and the final tensor amplification are all used.

A bare approximate reference-dimension cutoff, without an effective fixed-reference optimization routine, is a separate question. Fixed-reference faithful state pairs can still be arbitrarily ill conditioned. The argument above therefore makes no unqualified approximate-reference-cutoff claim. It does exclude a cutoff together with a total fixed-reference approximation procedure sufficient to approximate $\mathcal{V}$ at every requested accuracy.

15 Scope and relation to existing frameworks

The reduction uses generic complex POVMs and opposite Pauli chirality. It does not transfer automatically to real or transpose-compatible denominator tuples. Finite cutoffs or lift characterizations for narrower classes are compatible with the theorem.

The admissible inverse kernels always act on a finite-dimensional *spatial two-factor* domain. A compression that mixes the factors can destroy that domain. Results for free spectrahedra or unconstrained matrix evaluations, such as the framework in [4], therefore cannot be used to infer a cutoff here without proving that the spatial domain is preserved.

Complete channel orders and contraction-coefficient frameworks are developed in [2]; estimation and complete information comparisons are discussed in [3]. These are contextual references, not substitutes for any of the reduction lemmas above. This note makes no claim of exhaustive novelty verification. The exact theorem concerns qualitative preorder membership; the additive theorem concerns the bounded entropy defect $\mathcal{V}$. Neither identifies numerical contraction coefficients, covers all restricted-marginal variants, or asserts a common Hermiticity-preserving lift characterization for generic measurement pairs.

Every indispensable new step of the reduction has been included: the honest domain realization, augmented-anchor estimates, local smoothing, question normalization, finite gapped penalty, ordered inverse compiler, rational complex numerator implementation, faithful slice embedding, and uniform global completion. The additive strengthening also includes the Schmidt-sector argument, dimension-independent faithful normalization, explicit entropy-filter margin, and finite tensor amplification. The imported nonlocal-game complexity theorem remains the external premise. The note’s research status remains AI-generated and not peer-reviewed.

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